

EQUIVARIANT RIGIDITY OF LOCALIZED BOST–CONNES C^* -DYNAMICAL SYSTEMS: THE FIXED-POINT ALGEBRA IS THE CARTAN SUBALGEBRA

RUQING CHEN

ABSTRACT. By Paper IX of the series the von Neumann algebra of a localized Bost–Connes system remembers nothing: it is injective, hence determined by its flow of weights. There it was also observed that the obstruction group Ξ_β survives at the C^* -level, through the KMS_β simplex. We make that precise and determine how much of the arithmetic an equivariant isomorphism of C^* -dynamical systems recovers.

The technical point on which everything rests is elementary and, as far as we can tell, has not been isolated. Renault’s reconstruction theorem applies to Cartan *pairs*, whereas an isomorphism of C^* -dynamical systems is only required to intertwine the flows; Cartan subalgebras of a C^* -algebra are in general far from unique, so the passage from one to the other is not automatic. We show it is automatic here, for a reason internal to the system:

$$\mathcal{A}_{K,S}^\sigma = \overline{\text{span}}\{\mu_a f \mu_b^* : \mathbf{N}a = \mathbf{N}b\} = C(Y_S) \quad \text{whenever } \mathbf{N} : J_S \rightarrow \mathbb{N} \text{ is injective,}$$

since $\mathbf{N}a = \mathbf{N}b$ then forces $\mathbf{a} = \mathbf{b}$ and $\mu_a f \mu_a^* = \alpha_a(f) \in C(Y_S)$. The Cartan subalgebra is therefore *canonically the fixed-point algebra of the flow*, and every σ -equivariant isomorphism preserves it. Norm-separation is the same thing as $\mathbb{Q}$ -linear independence of the $\log \mathbf{N}\mathbf{p}$, holds exactly when S contains at most one prime above each rational prime, and is automatic over $\mathbb{Q}$.

With that, Renault’s theorem gives: a σ -equivariant isomorphism $\mathcal{A}_{K_1,S_1} \cong \mathcal{A}_{K_2,S_2}$ induces an isomorphism of the groupoids of the two systems intertwining the scaling cocycles $c(\mathbf{a}) = \log \mathbf{N}a$. The full bisections on which the cocycle is constant are exactly the semigroup elements, so the isomorphism induces a norm-preserving bijection $S_1 \rightarrow S_2$: the multiset $\{\mathbf{N}\mathbf{p}\}_{\mathbf{p} \in S}$ is recovered. The symmetry group G_S acts by automorphisms of the groupoid fixing every semigroup bisection, and over $\mathbb{Q}$, for sets of primes that are Dirichlet-dense at each of their members, these are the only such automorphisms; whenever the isomorphism intertwines the symmetry actions — automatic in that case — it matches the Frobenius classes and transports the whole chain $\{\Xi_\beta\}$. In general what is recovered is the compact space G_S , the KMS_β simplices, the transition locus and β_c ; whether the group structure of G_S is an invariant of the dynamical system for fields with infinite unit group we do not know.

We are careful about what is *not* recovered. The system sees only the triple $(S, \mathbf{N}|_S, G_S)$ with its Frobenius data; the field K enters only through that triple, and whether the triple determines K is an arithmetic question we do not settle and which, for $S = P_K$, belongs to the literature on Bost–Connes reconstruction. The narrow class number is not recovered by any argument here. The KMS bundle is a weaker invariant than the groupoid: it sees the simplices and their symmetry, but we see no way to extract the individual norms from it. Finally, the proposed equivariant classification of the Kirchberg boundary by $KK_{\mathbb{R}}$ is not available to us: the classification of flows on Kirchberg algebras up to cocycle conjugacy is not a settled theory, and we do not assert it.

1. THE FIXED-POINT ALGEBRA

We use the notation of [6]: $\mathcal{A}_{K,S} = C(Y_S) \rtimes J_S$, spanned by $\mu_a f \mu_b^*$, with $\sigma_t(\mu_a) = \mathbf{N}a^{it} \mu_a$ and $\alpha_a(f) = \mu_a f \mu_a^*$; it is the unital full corner of the groupoid algebra of $\mathcal{G}_S = I_S \rtimes \tilde{Y}_S$, and we write

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XIII-equivariant/](#).

$\mathcal{G}_S|_{Y_S}$ for the reduction of $\mathcal{G}_S$ to the compact open subset Y_S of its unit space, an étale groupoid with $C^*(\mathcal{G}_S|_{Y_S}) = \mathcal{A}_{K,S}$ and $C(Y_S)$ as diagonal.

The series. The foundational paper is [6] and Paper I is [7]; Papers II, III and IX, cited by numeral, are in preparation, and the facts used from them are re-proved here.

Definition 1.1. Say that S is *norm-separated* if $N : J_S \rightarrow \mathbb{N}$, $\mathfrak{a} \mapsto N\mathfrak{a}$, is injective.

Lemma 1.2 (When norm-separation holds). *The following are equivalent:*

- (1) S is norm-separated;
- (2) the numbers $\log N\mathfrak{p}$, $\mathfrak{p} \in S$, are linearly independent over $\mathbb{Q}$;
- (3) S contains at most one prime above each rational prime.

In particular every S is norm-separated when $K = \mathbb{Q}$, and S fails to be norm-separated as soon as it contains two primes above the same rational prime — two primes of equal norm, or a prime together with another whose norm is a power of its residue characteristic.

Proof. (1) $\Leftrightarrow$ (2): J_S is free on S , so $N\mathfrak{a} = N\mathfrak{b}$ with $\mathfrak{a} \neq \mathfrak{b}$ is a nontrivial relation $\sum_{\mathfrak{p}} (a_{\mathfrak{p}} - b_{\mathfrak{p}}) \log N\mathfrak{p} = 0$ with integer coefficients, and every integer relation arises this way; integer and rational independence agree. (2) $\Leftrightarrow$ (3): $\log N\mathfrak{p} = f_{\mathfrak{p}} \log p$ for $\mathfrak{p}$ above the rational prime p with residue degree $f_{\mathfrak{p}}$, and the $\log p$ are $\mathbb{Q}$ -independent by unique factorization; so a rational relation among the $\log N\mathfrak{p}$ decomposes into relations among the primes above each fixed p , and two primes above the same p satisfy $f_{\mathfrak{p}'} \log N\mathfrak{p} - f_{\mathfrak{p}} \log N\mathfrak{p}' = 0$. Compare [7, Lem. 4.1]. $\square$

Example 1.3. For $K = \mathbb{Q}(i)$: the two primes above 5 both have norm 5, so any S containing both fails; a set containing one prime above each of some rational primes — one of the two primes above each $p \equiv 1 \pmod{4}$, inert primes $p \equiv 3 \pmod{4}$ with norm p^2 , the prime $1+i$ — is norm-separated. For $K = \mathbb{Q}(\sqrt{2})$ the set of all primes fails at every split prime. The failure by equal norms is the generic one.

Theorem 1.4 (The Cartan subalgebra is the fixed-point algebra). *If S is norm-separated then*

$$(1.1) \quad \mathcal{A}_{K,S}^{\sigma} := \{x \in \mathcal{A}_{K,S} : \sigma_t(x) = x \ \forall t\} = C(Y_S).$$

Consequently the Cartan subalgebra $C(Y_S) \subset \mathcal{A}_{K,S}$ is canonically determined by the flow, and every σ -equivariant isomorphism $\Phi : \mathcal{A}_{K_1,S_1} \rightarrow \mathcal{A}_{K_2,S_2}$ satisfies $\Phi(C(Y_{S_1})) = C(Y_{S_2})$, hence is an isomorphism of Cartan pairs.

Proof. A spectral subspace argument: $\sigma_t(\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*) = (N\mathfrak{a}/N\mathfrak{b})^{it} \mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*$, so the elements $\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*$ are eigenvectors and σ is the restriction to $\mathbb{R}$ of an action of the compact dual of the countable group $\Gamma_{\beta} = \langle N\mathfrak{p} \rangle \leq \mathbb{R}_{>0}$, in which $\mathbb{R}$ is dense since Γ_{β} is torsion free; averaging over that compact group is a conditional expectation onto the fixed-point algebra which kills every eigenvector of nonzero weight, so the fixed-point algebra is the closed span of those $\mu_{\mathfrak{a}} f \mu_{\mathfrak{b}}^*$ with $N\mathfrak{a} = N\mathfrak{b}$. Norm-separation gives $\mathfrak{a} = \mathfrak{b}$, and then $\mu_{\mathfrak{a}} f \mu_{\mathfrak{a}}^* = \alpha_{\mathfrak{a}}(f)$, a continuous function on Y_S supported on the clopen set $\mathfrak{a}Y_S$. Hence $\mathcal{A}^{\sigma} \subseteq C(Y_S)$; the reverse inclusion holds because σ_t fixes $C(Y_S)$ pointwise. For the consequence, $\Phi \circ \sigma_t^1 = \sigma_t^2 \circ \Phi$ gives $\Phi(\mathcal{A}_1^{\sigma^1}) = \mathcal{A}_2^{\sigma^2}$. $\square$

Remark 1.5 (Why this needs saying). Renault's reconstruction theorem is a statement about Cartan pairs $(B \subset A)$: it recovers the twisted groupoid from the pair, not from A alone. A C^* -algebra may carry many non-conjugate Cartan subalgebras, so an isomorphism of algebras — even an equivariant one — carries no information about groupoids unless it is known to respect the pair. Theorem 1.4 supplies exactly that, by identifying the Cartan subalgebra intrinsically, in terms of the flow that the isomorphism is assumed to intertwine. Without it the reconstruction argument of §2 has a gap.

Remark 1.6 (The unseparated case). If S is not norm-separated then $\mathcal{A}^\sigma$ is strictly larger than $C(Y_S)$: it contains $\mu_a f \mu_b^*$ for distinct a, b of equal norm, which is not in the diagonal. The fixed-point algebra is then non-commutative and Theorem 1.4 fails. This does not show that the reconstruction fails, only that this route to it does; a different intrinsic characterization of $C(Y_S)$ would be needed, and we do not have one.

2. RECONSTRUCTION

Throughout this section S is norm-separated. We first record what in the groupoid $\mathcal{G}_S|_{Y_S}$ is intrinsic. Its arrows are the pairs (ab^{-1}, y) with $y \in Y_S$ and $ab^{-1}y \in Y_S$, the scaling cocycle is $c(ab^{-1}, y) = \log Na - \log Nb$, and for $a \in J_S$ the set $B_a = \{(a, y) : y \in Y_S\}$ is a full bisection — an open set of arrows on which the source map is a homeomorphism onto the whole unit space — with range aY_S and $c \equiv \log Na$.

Lemma 2.1 (The semigroup is intrinsic). *The full bisections of $\mathcal{G}_S|_{Y_S}$ on which c is constant are exactly the B_a , $a \in J_S$, and $B_a B_b = B_{ab}$. Consequently an isomorphism of étale groupoids $\theta : \mathcal{G}_{S_1}|_{Y_{S_1}} \rightarrow \mathcal{G}_{S_2}|_{Y_{S_2}}$ with $c_2 \circ \theta = c_1$ induces a monoid isomorphism $\pi : J_{S_1} \rightarrow J_{S_2}$ with $N\pi(a) = Na$, hence a bijection $S_1 \rightarrow S_2$ preserving norms.*

Proof. By Lemma 1.2, norm-separation means that $N : I_S \rightarrow \mathbb{Q}_{>0}$ is injective on the group of fractions $I_S = \mathbb{Z}^{(S)}$, so the value $c(\gamma)$ determines the element $g \in I_S$ with $\gamma = (g, y)$. A full bisection contains for each $y \in Y_S$ exactly one arrow with source y ; if c is constant on it, all these arrows have the same $g = ab^{-1}$ with $a, b \in J_S$ coprime, and the arrow (ab^{-1}, y) exists in $\mathcal{G}_S|_{Y_S}$ only for $y \in bY_S$. Fullness forces $bY_S = Y_S$, i.e. $b = 1$, so the bisection is B_a . The composition rule is $(a, by)(b, y) = (ab, y)$. An isomorphism θ preserving c maps full bisections with constant c to full bisections with the same constant, and preserves composition, so $B_a \mapsto B_{\pi(a)}$ defines a monoid isomorphism of the free abelian monoids J_{S_i} with $N\pi(a) = Na$; a monoid isomorphism of free abelian monoids maps the free generators to the free generators. $\square$

Definition 2.2. Let $\text{Aut}_0(\mathcal{G}_S|_{Y_S})$ be the group of automorphisms of the topological groupoid $\mathcal{G}_S|_{Y_S}$ that map each bisection B_a onto itself, with the topology of uniform convergence on Y_S ; equivalently, the automorphisms of $\mathcal{A}_{K,S}$ fixing every μ_a and preserving $C(Y_S)$. The translation action of G_S on $Y_S = \widehat{\mathcal{O}}_S \times_{U_S} G_S$ commutes with the partial action of J_S , so it gives an injective continuous homomorphism $G_S \rightarrow \text{Aut}_0(\mathcal{G}_S|_{Y_S})$. For $K = \mathbb{Q}$ say that S is *Dirichlet-dense* if for every $p \in S$ the primes of $S \setminus \{p\}$ generate a dense subgroup of $\mathbb{Z}_p^\times$, i.e. meet every residue class modulo every power of p ; the set of all primes, and any set of density one in every arithmetic progression, is Dirichlet-dense.

Proposition 2.3 (The symmetry group is intrinsic over $\mathbb{Q}$). *Let $K = \mathbb{Q}$ and let S be Dirichlet-dense. Then $G_S \rightarrow \text{Aut}_0(\mathcal{G}_S|_{Y_S})$ is an isomorphism.*

Proof. For $K = \mathbb{Q}$, $Y_S = \prod_{p \in S} \mathbb{Z}_p$ with $a \in J_S$ acting by multiplication and $G_S = \prod_{p \in S} \mathbb{Z}_p^\times$ acting by multiplication as well. Let $\psi \in \text{Aut}_0$. It commutes with multiplication by every $n \in J_S$ and preserves the ranges nY_S , hence the valuations, hence the set $G_S = Y_S^\times$ of units; write $\psi(u) = u c(u)$ for $u \in G_S$, with $c : G_S \rightarrow G_S$ continuous. For $x \in Y_S$ with no zero component, $x = nu$ uniquely with $n \in J_S$, $u \in G_S$, and $\psi(x) = n\psi(u) = x c(u)$. Let $p \in S$ and consider points x with $x_p = 0$: they are limits of $n_k u_k$ with $n_k \in J_S$ divisible by p^k , and the p -components $u_{k,p}$ of the units are unconstrained, since $(n_k)_p \rightarrow 0$; so continuity of ψ at such points forces the components $c(u)_q$, $q \neq p$, not to depend on u_p . Varying p over $S \setminus \{q\}$, the component $c(u)_q$ depends only on u_q : $c(u) = (c_q(u_q))_q$ with $c_q : \mathbb{Z}_q^\times \rightarrow \mathbb{Z}_q^\times$ continuous. Now compare $\psi(nx)$ and $n\psi(x)$ for x a unit and $n \in J_S$: the q -component of nx is $q^{v_q(n)} n'_q x_q$ with $n'_q = n/q^{v_q(n)}$, so $\psi(nx)_q = q^{v_q(n)} n'_q x_q c_q(n'_q x_q)$, while $(n\psi(x))_q = q^{v_q(n)} n'_q x_q c_q(x_q)$. Hence $c_q(n'_q x_q) = c_q(x_q)$ for all

$n \in J_S$ and $x_q \in \mathbb{Z}_q^\times$: c_q is invariant under multiplication by the unit parts of the elements of J_S , which generate the subgroup of $\mathbb{Z}_q^\times$ generated by the primes of $S \setminus \{q\}$, dense by hypothesis. So each c_q is constant, c is a constant $g \in G_S$, and ψ is multiplication by g . $\square$

Remark 2.4. For a general number field the same argument gives less. The constraint from continuity at the points with infinite valuation at $\mathfrak{p}$ concerns ψ only modulo the image $r(\mathcal{O}_{\mathfrak{p}}^\times) \subseteq G_S$ of the local units, and for fields with infinite unit group the images of the local units at different primes are not independent in G_S , the global units sitting diagonally in all of them. We do not know whether $\text{Aut}_0(\mathcal{G}_S|_{Y_S}) = G_S$ in general; the compact model $\overline{\mathbb{N}}^S \times G_S$ used elsewhere in the series differs from Y_S exactly at these points, and would give the equality for every Frobenius-dense S , but it is not the space on which $\mathcal{A}_{K,S}$ is built.

Theorem 2.5 (Equivariant reconstruction). *Let S_1, S_2 be infinite and norm-separated, and suppose $\Phi : (\mathcal{A}_{K_1, S_1}, \sigma^1) \rightarrow (\mathcal{A}_{K_2, S_2}, \sigma^2)$ is an isomorphism of C^* -dynamical systems. Then:*

- (1) Φ is an isomorphism of Cartan pairs and, by Renault's theorem, is induced by an isomorphism of étale groupoids $\theta : \mathcal{G}_{S_1}|_{Y_{S_1}} \rightarrow \mathcal{G}_{S_2}|_{Y_{S_2}}$;
- (2) θ intertwines the scaling cocycles, $c_2 \circ \theta = c_1$;
- (3) θ induces a norm-preserving bijection $\pi : S_1 \rightarrow S_2$, so the multisets $\{\mathbf{N}\mathfrak{p}\}_{\mathfrak{p} \in S_1}$ and $\{\mathbf{N}\mathfrak{p}\}_{\mathfrak{p} \in S_2}$ coincide;
- (4) θ maps the valuation fibres of Y_{S_1} homeomorphically onto those of Y_{S_2} , compatibly with π ; in particular the fibre over 0, which is the compact space G_{S_1} , is carried homeomorphically onto G_{S_2} ;
- (5) Φ induces affine isomorphisms $\text{KMS}_\beta(\mathcal{A}_{K_1, S_1}) \cong \text{KMS}_\beta(\mathcal{A}_{K_2, S_2})$ for every β , so the compact spaces $G_{S_1}/\Xi_\beta(K_1, S_1)^\perp$ and $G_{S_2}/\Xi_\beta(K_2, S_2)^\perp$ are homeomorphic, $|\Xi_\beta|$ agree when finite, the transition loci agree, and $\beta_c(S_1) = \beta_c(S_2)$;
- (6) suppose Φ intertwines the symmetry actions, $\Phi \circ \tau_g = \tau_{\iota(g)} \circ \Phi$ for an isomorphism of topological groups $\iota : G_{S_1} \rightarrow G_{S_2}$ — which is automatic, with $\iota = \theta(\cdot)\theta^{-1}$ on Aut_0 , when $K_1 = K_2 = \mathbb{Q}$ and the S_i are Dirichlet-dense, by Proposition 2.3. Then $\iota(\sigma_{\mathfrak{p}}) \in \sigma_{\pi(\mathfrak{p})} r(\mathcal{O}_{\pi(\mathfrak{p})}^\times)$ for every $\mathfrak{p} \in S_1$: the Frobenius classes modulo inertia are matched, and ι^{-1} maps $\Xi_\beta(K_2, S_2)$ onto $\Xi_\beta(K_1, S_1)$ for every $\beta > 0$.

Proof. (1) Theorem 1.4 makes Φ an isomorphism of Cartan pairs. The groupoid $\mathcal{G}_S|_{Y_S}$ is étale, Hausdorff, second countable, amenable, and topologically principal: the isotropy at a point with all valuations finite is trivial, since $\mathbf{a}\mathbf{b}^{-1}y = y$ forces $\mathbf{a} = \mathbf{b}$ on such points, and these points are dense. So $(C(Y_S) \subset \mathcal{A}_{K,S})$ is a Cartan pair with trivial Weyl twist, and Renault's theorem [4] gives θ inducing Φ . (2) Under $\Phi = \theta_*$ the flow is $\sigma_t(f)(\gamma) = e^{itc(\gamma)}f(\gamma)$ on functions supported on bisections, so $\Phi\sigma_t^1 = \sigma_t^2\Phi$ gives $e^{itc_2(\theta\gamma)} = e^{itc_1(\gamma)}$ for all t . (3) is Lemma 2.1. (4) θ maps $B_{\mathbf{a}}$ to $B_{\pi(\mathbf{a})}$, hence the ranges $\mathbf{a}Y_{S_1}$ to $\pi(\mathbf{a})Y_{S_2}$, hence the valuation fibres to the valuation fibres; the fibre over 0 is $U_S \times_{U_S} G_S = G_S$. (5) An isomorphism of C^* -dynamical systems carries KMS_β states to KMS_β states; the simplices are $\text{Prob}(G_S/\Xi_\beta^\perp)$ by [6, Thm. 3.6], and an affine isomorphism of Bauer simplices is a homeomorphism of their extreme boundaries; the transition locus is the set of β at which the simplex changes. (6) Since $\Phi = \theta_*$ preserves the Cartan pair, intertwining the symmetry actions means $\theta \circ \tau_g = \tau_{\iota(g)} \circ \theta$ on Y_{S_1} ; when $\text{Aut}_0 = G_S$ on both sides this holds with $\iota = \theta(\cdot)\theta^{-1}$, an isomorphism of topological groups since θ matches the bisections $B_{\mathbf{a}}$. On the valuation fibres, which are G_S -torsors, θ is ι -equivariant, so in the coordinates $y = (v, h)$ of [6, Lem. 2.10] on the points with finite valuations, $\theta(v, h) = (\pi(v), \iota(h)c_v)$ with constants $c_v \in G_{S_2}$, and $\theta(\mathbf{p} \cdot (v, h)) = \pi(\mathbf{p}) \cdot \theta(v, h)$ reads $\iota(\sigma_{\mathbf{p}})\iota(h)c_{v+e_{\mathbf{p}}} = \sigma_{\pi(\mathbf{p})}\iota(h)c_v$, i.e. $c_{v+e_{\mathbf{p}}} = u_{\mathbf{p}}c_v$ with $u_{\mathbf{p}} = \iota(\sigma_{\mathbf{p}})^{-1}\sigma_{\pi(\mathbf{p})}$. The points $(ke_{\mathbf{p}}, h)$ converge in Y_{S_1} as $k \rightarrow \infty$ modulo the inertia image $r(\mathcal{O}_{\mathbf{p}}^\times)$, and their images $(ke_{\pi(\mathbf{p})}, \iota(h)u_{\mathbf{p}}^k c_0)$ must converge modulo $r(\mathcal{O}_{\pi(\mathbf{p})}^\times)$; a sequence u^k converging in

the compact group $G_{S_2}/r(\mathcal{O}_{\pi(\mathfrak{p})}^\times)$ has u trivial there. So $u_{\mathfrak{p}} \in r(\mathcal{O}_{\pi(\mathfrak{p})}^\times)$. Finally Ξ_β is characterized by the series $\sum_{\mathfrak{p}} N\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2$ [6, (1.1)], whose convergence class is unchanged when $\sigma_{\mathfrak{p}}$ is altered by inertia elements at $\mathfrak{p}$ for all but finitely many $\mathfrak{p}$, the KMS simplex being an invariant of the system; so it is transported by π and $\hat{\iota}$. $\square$

Remark 2.6 (The hypotheses). Norm-separation enters twice, in Theorem 1.4 and in Lemma 2.1; by Lemma 1.2 it is the same as the $\mathbb{Q}$ -independence of the $\log N\mathfrak{p}$, so there is only one arithmetic condition on S in parts (1)–(5), and it is automatic over $\mathbb{Q}$. Part (6) needs the symmetry actions to be intertwined: without that, θ still matches the fibres and the simplices, but a homeomorphism of compact groups carries no group structure, and we do not know whether G_S with its Frobenius classes is an invariant of the dynamical system in general (Remark 2.4).

3. WHAT IS NOT RECOVERED

Remark 3.1 (The field). Theorem 2.5 recovers the pair $(S, N|_S)$, and, when the symmetry actions are intertwined, the triple $(S, N|_S, G_S)$ with its Frobenius data. It does *not* recover K . The system is built from that triple and nothing else, so two number fields yielding the same triple give isomorphic dynamical systems, whatever their other invariants. Whether the triple determines K is an arithmetic question, not an operator-algebraic one; for $S = P_K$ it is the subject of the literature on reconstruction from Bost–Connes systems, which we have not been able to consult here and against which any claim of field reconstruction should be checked. We make no such claim.

Remark 3.2 (Arithmetic equivalence is not obviously the right equivalence). Arithmetically equivalent fields share their Dedekind zeta function, hence the multiset of norms $\{N\mathfrak{p}\}$ for $S = P_K$. They need not share G_K , which involves the class group and the units, and G_S is part of the recovered data by Theorem 2.5(6). So the invariant recovered here is finer than the zeta function and coarser than K ; locating it exactly is open. We note that the narrow class number $h_S^+ = |\text{Cl}_S^+|$ is not recovered by any argument of this paper: Cl_S^+ is the quotient of G_S by the image of the local units, a finite-index open subgroup which nothing in the dynamical system singles out.

4. THE KMS BUNDLE IS A WEAKER INVARIANT

Definition 4.1. The *KMS bundle* of $(\mathcal{A}_{K,S}, \sigma)$ is $\bigsqcup_{\beta>0} \text{KMS}_\beta(\mathcal{A}_{K,S})$ over $(0, \infty)$, with the fibre-wise affine structure and the G_S -action. An isomorphism of KMS bundles is a fibrewise affine bijection over the identity of $(0, \infty)$ together with an isomorphism ι of the symmetry groups, the two being compatible.

Proposition 4.2 (What the bundle sees). *A fibrewise affine bijection of KMS bundles over the identity of $(0, \infty)$ determines the compact spaces $G_S/\Xi_\beta^\perp$ for every β , hence the transition locus, β_c , and $|\Xi_\beta|$ where finite. An isomorphism of KMS bundles in the sense of Definition 4.1 determines in addition the chain $\{\Xi_\beta\}_{\beta>0}$ as subgroups of $\widehat{G_S}$, since $\Xi_\beta^\perp$ is the stabilizer in G_S of any extreme point of the fibre at β .*

Proof. Each of these is read off [6, Thm. 3.6]: the fibre at β is $\text{Prob}(G_S/\Xi_\beta^\perp)$, on whose extreme points G_S acts transitively with stabilizer $\Xi_\beta^\perp$. $\square$

Remark 4.3 (What the bundle does not see). We see no way to extract the individual norms $N\mathfrak{p}$ from the bundle. The fibres depend on S only through the sets $\{\mathfrak{p} : \chi(\sigma_{\mathfrak{p}}) \neq 1\}$ and their convergence exponents; two systems whose primes have different norms but the same detecting sets and the same convergence behaviour would have isomorphic bundles. The bundle is therefore

weaker than the groupoid of Theorem 2.5, which recovers the norms outright, and the chain of implications runs

equivariant C^* -isomorphism $\implies$ groupoid isomorphism $\implies$ KMS bundle isomorphism,

with the converses unproved. We do not claim the bundle is a complete invariant.

5. THE KIRCHBERG BOUNDARY: A CAUTION

Remark 5.1 (Flows on Kirchberg algebras). By Paper II of the series the boundary $\partial\mathcal{A}_{K,S}^{\text{aff}}$ of the affine system is a unital Kirchberg algebra satisfying the UCT, and it carries the flow $\sigma_t(\mu_{(a,p)}) = \text{Np}^{it}\mu_{(a,p)}$. It is tempting to classify the pair by equivariant KK . We do not, for two reasons.

First, the classification of *flows* — actions of $\mathbb{R}$ — on Kirchberg algebras up to cocycle conjugacy is not a settled theory. The Kirchberg–Phillips theorem [1] classifies the algebras; the equivariant theory is well developed for finite and, more recently, for many amenable *discrete* groups [5], but $\mathbb{R}$ -actions on purely infinite algebras remain substantially open, and we are not able to invoke a classification theorem here. (For the diagonal-preserving isomorphisms of §2, by contrast, the relevant rigidity is that of continuous orbit equivalence [2, 3], whose groupoid form is Renault’s theorem.)

Second, and more concretely, by Paper II this system has KMS_β states only at $\beta = 1$. Whatever the classification of the flow, the equilibrium invariant that carries the arithmetic in the Toeplitz system has collapsed to a single point on the boundary, so the boundary is not where the reconstruction of §2 can be carried out.

6. ASSESSMENT

Statement	Status	Reference
$\mathcal{A}^\sigma = C(Y_S)$ for norm-separated S	true	Thm. 1.4
norm-separation automatic over $\mathbb{Q}$	true	Lem. 1.2
equivariant iso preserves the Cartan pair	true	Thm. 1.4
Renault: pair $\Rightarrow$ groupoid	true	Thm. 2.5(1)
norm-separation = $\mathbb{Q}$ -indep. of $\log \text{Np}$	true	Lem. 1.2
norms $\{\text{Np}\}$ recovered	true	Thm. 2.5(3)
simplices, transition locus, β_c recovered	true	Thm. 2.5(5)
G_S , σ_p , $\{\Xi_\beta\}$ recovered	if symmetries intertwined	Thm. 2.5(6)
$\text{Aut}_0 = G_S$	true over $\mathbb{Q}$, Dirichlet-dense S	Prop. 2.3
h_S^+ recovered	not claimed	Rem. 3.2
the field K recovered	not claimed	Rem. 3.1
KMS bundle a complete invariant	not claimed; weaker	Rem. 4.3
Kirchberg boundary classified by $KK_{\mathbb{R}}$	not available	Rem. 5.1

The contrast with Paper IX is the point. There the von Neumann algebra was shown to retain nothing, the groupoid being amenable and Connes’ classification leaving only the flow of weights. Here the C^* -algebra *with its flow* retains the norms, the KMS simplices, and — once the symmetries are intertwined, which over $\mathbb{Q}$ is automatic — the symmetry group with its Frobenius classes and the whole chain $\{\Xi_\beta\}$. The mechanism is Theorem 1.4: the flow itself pins down the Cartan subalgebra, so the reconstruction theory of Renault becomes available to a merely equivariant isomorphism. Weak closure destroys the flow — the modular group of a single extremal state is an inner datum — and with it the pin.

Four questions. Is $C(Y_S)$ intrinsically characterized when S is not norm-separated, so that Theorem 2.5 extends beyond Lemma 1.2? Is $\text{Aut}_0(\mathcal{G}_S|_{Y_S}) = G_S$ for fields with infinite unit group,

so that the group structure of G_S is an invariant of the dynamical system there too? Is the KMS bundle a complete invariant after all, or is there a pair of systems with isomorphic bundles and non-isomorphic groupoids — we would expect the latter and suggest looking at sets with the same detecting data but different norms. And what exactly is the arithmetic content of the recovered triple $(S, N|_S, G_S)$: it is finer than the Dedekind zeta function, since G_S carries class group and unit data, and coarser than K ; where it sits between them we do not know.

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GUT GEOSERVICE INC., MONTREAL, CANADA

Email address: ruqing@hotmail.com